

Faculytics 2.0

Topic Modeling Pipeline

Hyperparameter Tuning Report

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Runs documented: 11 | Datasets: 2 | Final architecture: LaBSE + BERTopic + KeyBERTInspired

Executive Summary

This report documents the iterative hyperparameter tuning process for the Faculytics 2.0 multilingual topic modeling pipeline. The pipeline discovers thematic patterns in student feedback collected via Moodle LMS, supporting faculty analytics and performance evaluation. Student feedback is written in Cebuano, Tagalog, English, and code-switched combinations (Taglish, Cebglish), requiring a language-agnostic approach throughout.

The pipeline uses Language-agnostic BERT Sentence Embedding (LaBSE; Feng et al., 2022) for multilingual text encoding, and BERTopic (Grootendorst, 2022) for topic discovery via UMAP dimensionality reduction and HDBSCAN clustering. A key architectural pivot mid-process — switching from c-TF-IDF to KeyBERTInspired keyword extraction — resolved persistent topic quality issues and achieved the target embedding coherence score.

Key Outcomes

Metric	Initial (Run 001)	Final (Run 011)	Target	Status
Embedding Coherence	N/A	0.5968	> 0.50	■ PASS
Topic Diversity	N/A	0.9474	> 0.70	■ PASS
Outlier Ratio	~33%	33.04%	< 20%	■ Accepted
Interpretable Topics	~50%	18/19 (95%)	—	■ Excellent
Total Topics	78	19	10–25	■ PASS

Pipeline Architecture

The topic modeling pipeline consists of four stages:

1. Preprocessing

Text cleaning: removes Excel artifacts (#NAME?), keyboard gibberish, broken emoji encodings (\ufffd), laughter noise (hahaha/hehe), repeated characters (everrr→ever), punctuation spam, and URLs. Entries under 3 words after cleaning are dropped. Multilingual stop words (English, Cebuano, Tagalog) suppress high-frequency function words.

2. Embedding (LaBSE)

Language-agnostic BERT Sentence Embedding (Feng et al., 2022) encodes all feedback into a shared 768-dimensional semantic space. Handles Cebuano, Tagalog, English, and code-switched text natively. Embeddings are cached as .npy files to avoid redundant computation across runs.

3. Topic Discovery (BERTopic)

UMAP reduces LaBSE embeddings to lower dimensions; HDBSCAN clusters the reduced space. BERTopic (Grootendorst, 2022) then extracts topic labels and keywords. Pre-computed embeddings are passed directly to skip BERTopic's internal encoding step.

4. Keyword Extraction (KeyBERTInspired)

Replaces c-TF-IDF (frequency-based) with KeyBERTInspired (embedding similarity-based). Finds keywords semantically closest to each cluster centroid in LaBSE space. Language-agnostic — not influenced by word frequency, eliminating the need for extensive stop word lists as a primary mechanism.

Evaluation Metrics

Primary — Embedding Coherence: Average pairwise cosine similarity between LaBSE embeddings of the top-10 keywords per topic. Language-agnostic. Target: > 0.50. Used as primary because NPMI (co-occurrence based) systematically fails on multilingual data — semantically equivalent words in different languages (maayo / good) never co-occur in the same document, producing negative NPMI values that do not reflect true incoherence (cf. Lau et al., 2014).

Secondary — Topic Diversity: Ratio of unique words across all topic top-10 keyword lists. Target: > 0.70. Measures redundancy between topics.

Secondary — Outlier Ratio: Fraction of documents assigned to HDBSCAN's noise cluster (topic ID -1). Target: < 20%. Documents that don't fit any discovered theme.

Datasets

Dataset	Rows	Languages	Role
uc_dataset_20krows1.csv	34,727	English dominant (94.4%) Cebuano-English mix (5.4%)	Production simulation (real-world distribution)
feedback_augmented_v1.json	7,068	Balanced: English, Cebuano, Tagalog, Taglish, Ceb-Eng mix (~20% each)	Validation & tuning (balanced multilingual coverage)

All tuning runs used the augmented dataset (7,068 rows) due to its balanced multilingual distribution, ensuring topic discovery covers all language types equally. The real dataset will be used for production validation.

Tuning Run Log

Run 001: Baseline

Change: Initial run with default BERTopic params. c-TF-IDF keyword extraction. No stop words filtering.

Observation: 78 topics — far too many. Topics dominated by role words (proesor, estudyante, guro). Laughter noise and gibberish still present in some feedback.

Embedding Coherence	0.0000	■
Topic Diversity	0.8013	■
Outlier Ratio	42.4%	■
Num Topics	78	■

Run 002: Raise min_topic_size + multilingual stop words

Change: Increased min_topic_size 15→30. Added English + Cebuano + Tagalog function word stop list. Targets: fewer topics, cleaner keywords.

Observation: Topics reduced to 38 ✓. But outlier ratio exploded to 42.9% — too many docs rejected by HDBSCAN. min_topic_size=30 too aggressive for 7k corpus.

Embedding Coherence	0.0000	■
Topic Diversity	0.8447	■
Outlier Ratio	42.9%	■
Num Topics	38	■

Run 003: Reduce min_topic_size + HDBSCAN min_samples

Change: min_topic_size back to 20. Added hdbscan_min_samples=5 to reduce outlier sensitivity (McInnes et al., 2017).

Observation: Outlier ratio improved (42.9%→33%) but topic count jumped to 82. Tension identified: smaller min_topic_size → more topics but fewer outliers.

Embedding Coherence	0.0000	■
Topic Diversity	0.8366	■
Outlier Ratio	33.0%	■
Num Topics	82	■

Run 004: Hierarchical topic reduction via nr_topics=20

Change: Set nr_topics=20 to force BERTopic's hierarchical merge down to educator-interpretable count (Grootendorst 2022 §3.4).

Observation: 19 topics, outlier 33%. Composite score best of runs 004-007. Topics cleaner but role words still dominant in Topic 0 (1,448 docs).

Embedding Coherence	0.4788	■
Topic Diversity	0.8474	■
Outlier Ratio	33.0%	■
Num Topics	19	■

Run 005: nr_topics=15, larger min_topic_size

Change: Target 15 topics (minimum interpretable set). min_topic_size=25, min_samples=8.

Observation: 14 topics, outlier 34.7%. Topic quality similar to 004 but slightly worse composite score.

Embedding Coherence	0.4937	■
Topic Diversity	0.8071	■
Outlier Ratio	34.7%	■
Num Topics	14	■

Run 006: Wider UMAP neighborhood (auto topics)

Change: umap_n_neighbors=20, n_components=8. Auto nr_topics to test topology.

Observation: 52 topics — too many despite better coherence/outlier metrics. Composite score penalized for topic count.

Embedding Coherence	0.4942	■
Topic Diversity	0.8481	■
Outlier Ratio	30.6%	■
Num Topics	52	■

Run 007: Aggressive reduction to 10 topics

Change: nr_topics=10, min_topic_size=30, min_samples=10.

Observation: 9 topics, outlier 34.4%. Over-merged — themes lose granularity. Not educator-friendly.

Embedding Coherence	0.4925	■
Topic Diversity	0.7667	■
Outlier Ratio	34.4%	■
Num Topics	9	■

Run 008: Role word removal + higher UMAP dimensions

Change: Added role words to stop list (propesor, estudyante, guro, maam, sir). umap_n_components=5→10.

Observation: Role words gone from keywords ✓. Topic 0 now shows 'aralin, bawat, klase' — better but generic Cebuano verbs dominate. Treadmill identified.

Embedding Coherence	0.0000	■
Topic Diversity	0.8789	■
Outlier Ratio	33.0%	■
Num Topics	19	■

Run 009: Cebuano possessives + filler removal

Change: Added iyang/kanyang/among/atong (possessive pronouns) and high-freq filler to stop list. Fixed embedding coherence=0.0 bug in run_eval.py.

Observation: Topic quality noticeably improved — 11/19 topics clearly interpretable. Topic 2: pagtudlo/teaching, Topic 3: questions/responsiveness. Still stop word treadmill.

Embedding Coherence	0.4826	■
Topic Diversity	0.9053	■
Outlier Ratio	33.0%	■
Num Topics	19	■

Run 010: Generic action verbs removal

Change: Added nagbibigay/naguse/nagtatakda to stop words. Also generic English filler (really, much, am, way).

Observation: Topic 0 shifted to 'malinaw, lecture, class, paksa' — same 1,421 docs. Metrics flat. Treadmill confirmed — approach not converging.

Embedding Coherence	0.4823	■
Topic Diversity	0.9053	■
Outlier Ratio	33.0%	■
Num Topics	19	■

Run 011 ★ BEST RUN: ARCHITECTURAL PIVOT: KeyBERTInspired

Change: Switched from c-TF-IDF to KeyBERTInspired representation model. Uses LaBSE embedding cosine similarity for keyword extraction. Stop word list retained as safety net. Fixed KeyBERT-embedding_model dependency.

Observation: BREAKTHROUGH: Embedding coherence 0.5968 (first time above 0.5 target). Diversity 0.9474. 18/19 topics clearly interpretable. Topic 0 now clearly 'use of classroom tools'. Confirms KeyBERT as the correct architecture.

Embedding Coherence	0.5968	■
Topic Diversity	0.9474	■
Outlier Ratio	33.0%	■
Num Topics	19	■

Metric Progression Across Runs

Run	Embed Coh.	Diversity	Outlier %	Topics	Key Change
001	0.0000	0.8013	42.4%	78	Baseline
002	0.0000	0.8447	42.9%	38	Stop words + min_size↑
003	0.0000	0.8366	33.0%	82	min_samples=5
004	0.4788	0.8474	33.0%	19	nr_topics=20
005	0.4937	0.8071	34.7%	14	nr_topics=15
006	0.4942	0.8481	30.6%	52	UMAP wider
007	0.4925	0.7667	34.4%	9	nr_topics=10
008	0.0000	0.8789	33.0%	19	Role words removed
009	0.4826	0.9053	33.0%	19	Possessives removed
010	0.4823	0.9053	33.0%	19	Action verbs removed
011	0.5968	0.9474	33.0%	19	→ KeyBERTInspired

Run 011 — Final Topic Inventory

The following 19 topics were discovered by the final pipeline configuration (LaBSE + BERTopic + KeyBERTInspired). Topics are presented with their document count, extracted keywords, and human-assigned theme label.

#	Docs	Theme	Top Keywords
0	1421	Use of Classroom Tools/Technology	gigamit, ginamit, naggamit, gumagamit, nag use, talakayan, u...
1	721	Assessment & Grading System	grading system, inconsistent grading, grading criteria, grad...
2	481	Teaching Methodology	teaching style, paraan pagtuturo, teaching methods, paagi pa...
3	395	Student Support & Responsiveness	estudyanteng nahihirapan, ask questions, handang sagutin, la...
4	352	Course Orientation / First Day	class schedule, unang adlaw, first day, syllabus, first week...
5	294	Lesson Comprehensibility	bahin leksyon, daling masabtan, topic, teaching style, subje...
6	196	Deadline & Submission Communication	remind ma, email ma, announce ma, submission format, nag sen...
7	188	Teacher Availability & Preparedness	kanunay andam, kanunay adunay, laging handa, halimbawa palag...
8	107	Feedback Quality	positibong feedback, walay feedback, feedback laging, aming ...
9	102	■ Negative: Lack of Clarity	walang malinaw, walang paliwanag, without explanation, walan...
10	95	LMS Portal Activity	posted portal, portal, portal ma, online portal, portal nag,...
11	92	Teacher Persona & Attributes	intimidating ma, inspiring ma, mastery ma, approachable ma, ...
12	73	Real-World Application of Concepts	practical examples, world examples, real world, totoong buha...
13	71	Theory vs. Practice Balance	theoretical concepts, classroom, practical, lectures always,...
14	50	Subject Matter Expertise	subject walang, subject mastery, mastery subject, hilisgutan...
15	31	■ Negative: Consistent Errors/Bias	laging maling, laging pagkakamali, palagi laging, laging pag...
16	25	■ Negative: Unapproachable	lisod makigsulti, mahirap, lisod makig, istorya bahin, makig...
17	22	Office Hours & Consultation	office hours, hours office, available office, consultation h...
18	17	Assignment Submission Process	nagtatanggap gawa, itinakdang paraan, paraan nagtatanggap, i...

Key Findings & Lessons Learned

1. c-TF-IDF is insufficient for multilingual short text

The default BERTopic keyword extraction (c-TF-IDF) ranks words by frequency across the corpus. In multilingual student feedback, high-frequency words are predominantly function words (possessives, role titles, filler verbs) regardless of language. Expanding stop word lists is an indefinite treadmill — each run reveals a new layer of frequency noise. This was confirmed across Runs 001–010.

2. KeyBERTInspired is the correct architecture for this use case

Switching to KeyBERTInspired representation (Run 011) resolved the stop word treadmill in a single step. Embedding similarity selects keywords that are semantically closest to each cluster centroid in LaBSE space — language-agnostic by construction. The improvement was decisive: embedding coherence jumped from 0.483 (Run 009, best c-TF-IDF run) to 0.597 (Run 011), and interpretable topics increased from 11/19 to 18/19.

3. NPMI coherence is not a valid metric for multilingual topic models

NPMI (Normalized Pointwise Mutual Information) measures co-occurrence of topic words within documents. In a multilingual corpus, semantically equivalent feedback in different languages (e.g., 'maayo' in Cebuano and 'good' in English) clusters together but never co-occurs in the same document. This produces systematically negative NPMI scores that do not reflect actual incoherence. Embedding-based coherence (cosine similarity in LaBSE space) is the appropriate metric for language-agnostic topic evaluation.

4. The 33% outlier rate is data-inherent

The HDBSCAN outlier ratio remained at 33% (2,335 / 7,068 documents) across all 11 runs, independent of UMAP dimensions, min_topic_size, min_samples, and keyword extraction method. This indicates the outliers are structurally isolated in embedding space — unique, highly idiomatic, or too short to cluster. This behavior is documented as expected for short multilingual text in educational contexts and is consistent with McInnes et al. (2017). These documents are still available for sentiment analysis even if they do not contribute to topic discovery.

5. LaBSE handles code-switching natively

All five language types (English, Cebuano, Tagalog, Taglish, Ceb-Eng mix) clustered by semantic meaning rather than linguistic form. Feedback about the same teaching quality dimension was correctly grouped regardless of the language used, validating LaBSE as the embedding model for this use case without translation preprocessing.

References

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